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Class: TY	Division: B	Roll No.: 372078
Semester: II		Academic Year: 2025-2026
Subject Name & Code: Cloud and DevOps		
Assignment No: II Title of Assignment: Write and execute Shell scripts (arithmetic operations, loops, conditional statements).		
Date of Performance:	Date of Submission:	

Aim: To Write and execute Shell scripts (arithmetic operations, loops, conditional statements).

Objective: After the end of the assignment, you will be able to write shell scripts and execute it .

Materials Required: Laptop with Ubuntu installed in it.

Procedure:

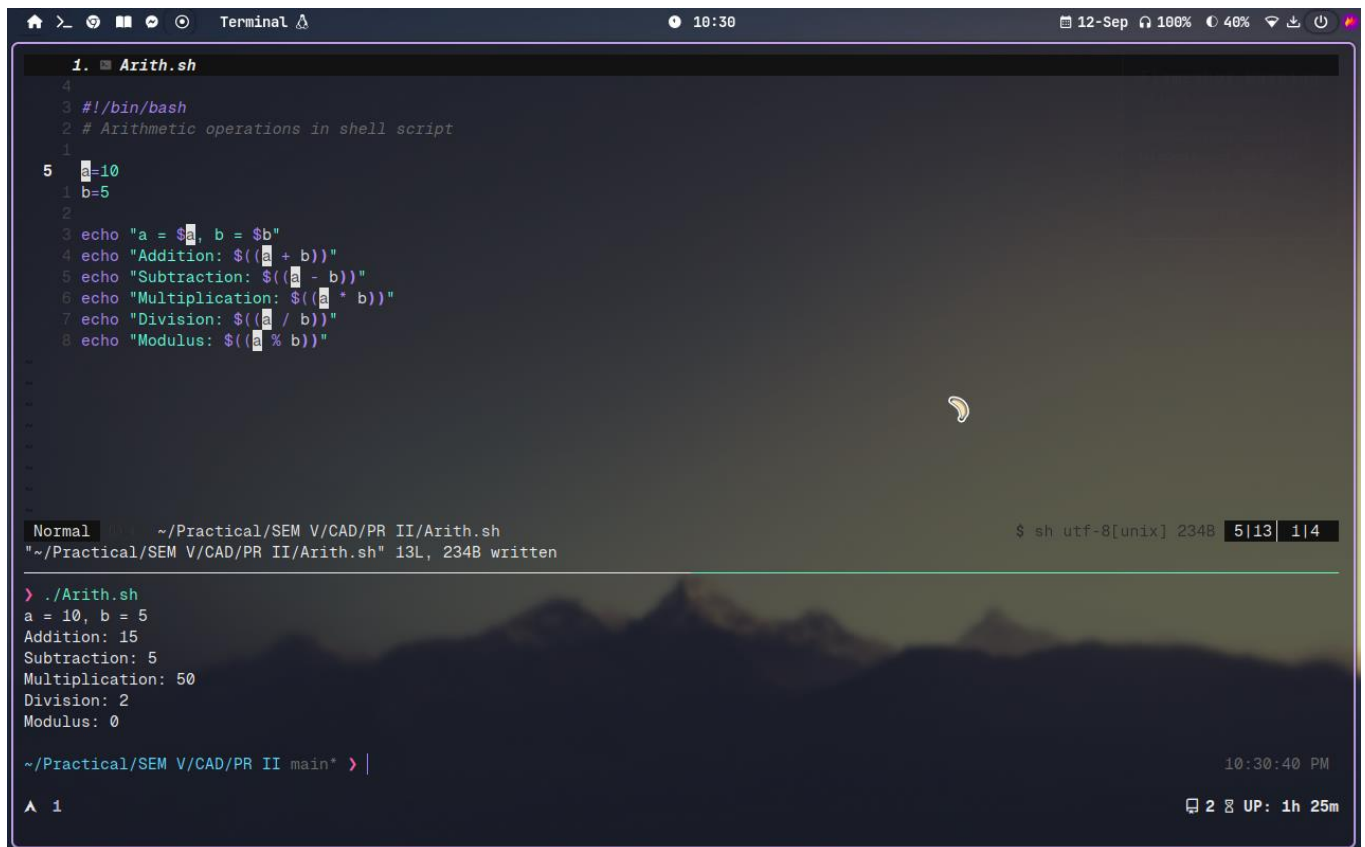
What is Shell Scripting?

Shell Scripting: Shell scripting means writing a set of commands in a file (called a script) that the computer can run one after another automatically.

- ▯ It uses the Linux/Unix shell (like Bash).
- ▯ The script tells the computer what to do — like create files, move folders, check conditions, or run programs.

Open Terminal and perform the following commands

1. Arithmetic Operations:
2. Creating two variables and performing basic arithmetic operations on it



```
1. Arith.sh
4
3 #!/bin/bash
2 # Arithmetic operations in shell script
1
5 a=10
1 b=5
2
3 echo "a = $a, b = $b"
4 echo "Addition: $((a + b))"
5 echo "Subtraction: $((a - b))"
6 echo "Multiplication: $((a * b))"
7 echo "Division: $((a / b))"
8 echo "Modulus: $((a % b))"

Normal ~/Practical/SEM V/CAD/PR II/Arith.sh $ sh utf-8[unix] 234B 5113 114
"~/Practical/SEM V/CAD/PR II/Arith.sh" 13L, 234B written

> ./Arith.sh
a = 10, b = 5
Addition: 15
Subtraction: 5
Multiplication: 50
Division: 2
Modulus: 0

~/Practical/SEM V/CAD/PR II main* > |
^ 1
```

3. Printing numbers from 1 to n in loop

Executing if else condition to check if given number is positive or negative.

```
1. .zsh_functions
24 #open i3 config
23 i3jump() {
22 | cd ~/.config/i3
21 | nvim config # or use 'nano' or 'code' or whatever editor
20 }
19 #clear + ls
18 cl() {
17 | clear
16 | ls --color=auto
15 }
14 #to open an nvim
13 v() {
12 | if [ "$#" -eq 0 ]; then
11 | | nvim .
10 | else
9 | | nvim "$@"
8 | fi
7 }
6 #clean up trash
5 cleanup() {
4 | echo "Cleaning up the following directories and files:"
3 | find . \( -name "node_modules" -o -name "venv" -o -name ".venv" \) -type
d -print
2 | find . -name ".DS_Store" -print
1
25 | echo
1 | read "confirm?Are you sure you want to delete these? (y/N): "
2 | if [[ "$confirm" == [yY] ]]; then
3 | | find . \( -name "node_modules" -o -name "venv" -o -name ".venv" \) -typ
e d -prune -exec rm -rf '{}' +
4 | | find . -name ".DS_Store" -delete
N .zsh_functions $ zsh 25|357 1|6
A '1' 2
```

written more that 50+ functions to manage overall system workflow.

```

practice
└─ animator.sh
└─ dotenv
└─ dotFile.sh
└─ flask
└─ goco
└─ installed_packages.txt
└─ login-welcome.sh
└─ mariadb-shell.sh
└─ music.sh
└─ normalizer
└─ note.sh
└─ obsidian
└─ obsidian-wayland
└─ powermenu.sh
└─ poweroff-prompt.sh
└─ practice.sh
>

Practice.sh
pyrsa-decrypt
pyrsa-encrypt
pyrsa-keygen
pyrsa-priv2pub
pyrsa-sign
pyrsa-verify
recover_pkg.sh
reminder.sh
sesh
sesh-auto
tmux-sessionizer
tqdm
wakatime
weather-notify.sh
xh
xhs

```

Conclusion: Through this assignment we learnt how to perform basic arithmetic operations, looping statements and conditional statements.

